Linear Regression using Boston Housing Dataset - Viva Questions & Answers

1. What is Linear Regression?
Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values.

2. Why did you use Linear Regression for this project?
Because house price is a continuous value, and Linear Regression works well for predicting continuous outputs.

3. What is the Boston Housing Dataset?
It is a famous dataset used in machine learning containing information about houses in Boston like crime rate, number of rooms, tax rate, etc., used to predict house prices.

4. What are the features in the dataset?
- CRIM → Crime rate
- RM → Average number of rooms
- AGE → Age of property
- TAX → Property tax rate
- PTRATIO → Student-teacher ratio
- LSTAT → Lower status population percentage

5. What is the target variable?
MEDV (Median value of owner-occupied homes) — the house price.

6. What is supervised learning?
A type of machine learning where input and output data are already labeled.

7. What is training data?
Data used to train the machine learning model.

8. What is testing data?
Data used to evaluate model performance on unseen data.

9. How did you divide the dataset?
Usually 70% or 80% for training and 20% or 30% for testing.

10. Why do we split the dataset?
To check whether the model performs well on unseen data and avoids overfitting.

11. What is overfitting?
When a model memorizes training data and performs poorly on new data.

12. What is underfitting?
When a model cannot learn patterns properly from data.

13. What is the formula of Linear Regression?
$y = mx + c$

Where:
- y = predicted value
- m = slope
- x = input feature

• c = intercept

For multiple features:
$y = b0 + b1x1 + b2x2 + ... + bnxn$

14. What libraries did you use in Python?
• pandas
• numpy
• matplotlib
• scikit-learn

15. What is pandas used for?
Pandas is used for data handling and analysis.

16. What is NumPy?
NumPy is a library used for numerical operations and arrays.

17. What is scikit-learn?
A Python library used for machine learning algorithms.

18. What evaluation metrics did you use?
• Mean Absolute Error (MAE)
• Mean Squared Error (MSE)
• R^2 Score

19. What is MAE?
Average absolute difference between actual and predicted values.

20. What is MSE?
Average squared difference between actual and predicted values.

21. What is R^2 Score?
It measures how well the model fits the data. Higher value means better performance.

22. What is prediction in machine learning?
Using trained data to estimate unknown outputs.

23. What is a model?
A mathematical representation trained using data for prediction.

24. Why is RM feature important?
Because houses with more rooms generally have higher prices.

25. What is feature selection?
Selecting important input variables for model training.

26. What is data preprocessing?
Cleaning and preparing data before training the model.

27. What is normalization?
Scaling data into a fixed range for better performance.

28. What are the applications of Linear Regression?

• House price prediction
• Sales forecasting
• Weather forecasting
• Stock analysis

29. What are the limitations of Linear Regression?

• Assumes linear relationship
• Sensitive to outliers
• Cannot model complex patterns well

30. Why is machine learning important?

Machine learning helps automate predictions and decision-making using data.